How Much Personal History Is Enough?

Measuring the data requirement for individualised financial behavioural anomaly detection

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A new customer opens an account

- Personal detector: compare with your own past
- Day one: your past is empty
- 900 units: normal for one, wild for another
- One population rule serves people badly

Where does the baseline start to work?

- Deployed systems filter "unseasoned users"
- The cutoff is assumed, not measured
- We found no published measurement

How many transactions are enough?

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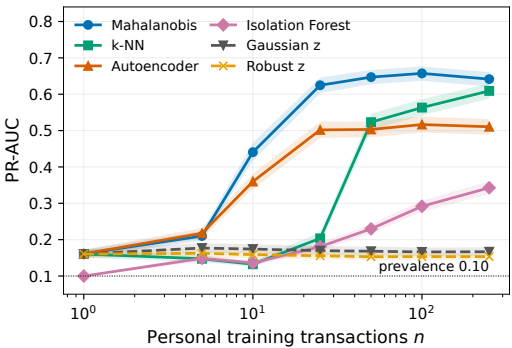
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- No detector's score defines the labels

Three data layers, six detectors

- Berka: 300 real bank accounts
- Simulation: 240 people, variability set
- TabFormer: 300 entities, synthetic
- Mahalanobis, k-NN, forest, autoencoder, z-scores

25 transactions reach 90% of best



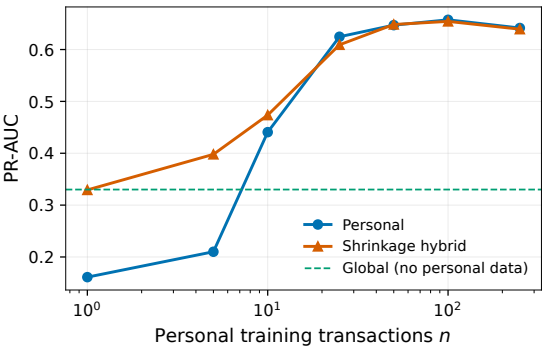
- Mahalanobis PR-AUC **0.657**
- 98% of the gain by n = 50
- Both z-scores never learn

The threshold depends on the definition

Detector	Rel-90	Rel-95
Mahalanobis	25	25
Autoencoder	25	25
k-NN	100	250
Isolation forest	250	250

- Same 25 at all **5** cohort seeds
- Stability: never met by a learner

Below n = 10, trust the crowd



- Population model: 400 other accounts
- Personal worse until n = 10
- k-NN: worse until n = 50

Shrink toward the crowd

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- Never meaningfully worse than either parent
- Fails if signal lives in the features

How you set the alert threshold matters

n	In-sample flag rate	LOO flag rate	LOO FPR
5	0.989	0.152	0.151
10	0.960	0.012	0.007
25	0.391	0.110	0.066
50	0.251	0.126	0.073

- n = 10: **96%** flagged in-sample

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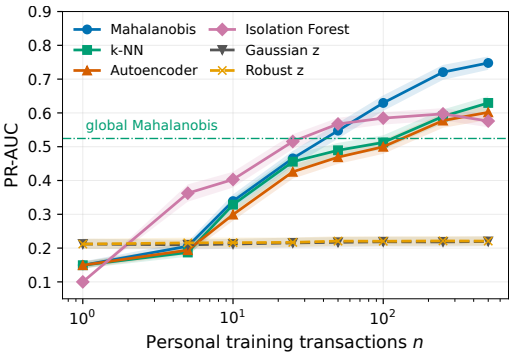
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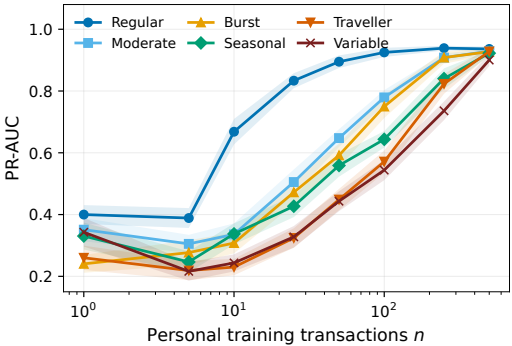
- $n = 10$: **96%** flagged in-sample
- Leave-one-out: **1.2%** flagged
- FPR 0.066 at $n = 25$ (nominal 0.05)

A second dataset reverses a ranking



- Same shape, later threshold (250)
- Isolation forest best at small n
- Worst detector on Berka

Variability sets how fast it learns



- Regular: 50 transactions; variable: 500
- Simulation: Spearman $\rho = 0.48$ to 0.71
- Real accounts: no descriptor predicts it

Recent history beats more history

- 50 to 250 transactions: $0.647 \rightarrow 0.642$
- Same 50 from $4\times$ span: -0.036
- Drift is real, but doesn't predict loss
- $\rho = -0.08$, Holm $p = 0.32$

Limits and ethics

- One bank, 1993 to 1998; second dataset synthetic
- 10% injection far above fraud prevalence
- A flag is a deviation, not fraud
- Audit is district-level only

Takeaways

- Baseline ranks well from about 25 transactions
- Hybrid removes the cold-start penalty
- Calibration moves the operating point
- Keep recent history, not all history

Questions?



Paper and code

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